

Deepak K

Electrical and Electronics Engineering undergraduate with experience in robotics, embedded systems, and autonomous systems

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EDUCATION

New Horizon College of Engineering

B.E. in Electrical and Electronics Engineering | CGPA: 9.24/10.0 (till 5th Sem)

- Relevant Coursework: Embedded Systems, Control Systems, Microcontrollers, Electronics Circuits, VLSI Design

Bengaluru, Karnataka

Aug 2023 – May 2027

St. Francis Composite PU College

Pre-University Course (PCME) | 92.17%

Bengaluru, Karnataka

Graduated 2022

TECHNICAL SKILLS

Programming: Arduino C, C, Python

Robotics & ROS 2: ROS 2 (Jazzy), micro-ROS, Navigation2, SLAM Toolbox, AMCL, URDF, Behavior Trees, TF Transforms

Autonomous Systems: LiDAR SLAM, Monte Carlo Localization, Occupancy Grid Mapping, Path Planning, Sensor Fusion

Computer Vision & AI: YOLOv8, OpenCV, Real-Time Object Detection, Deep Learning, Transfer Learning, Reinforcement Learning

Control & Embedded: PID, MPC, Inverse/Forward Kinematics, ESP32, Arduino, Raspberry Pi, Real-Time Systems

Hardware: 2D LiDAR, IMU, Encoders, IR Sensor Arrays, Ultrasonic Sensors, Motor Drivers

Tools: Gazebo, RViz, Git/GitHub, Linux (Ubuntu), Google Colab, CAD

TECHNICAL PROJECTS

LiDAR-Powered Autonomous Mapping & Navigation | ROS 2, SLAM Toolbox, AMCL, Nav2, Gazebo | [GitHub](#)

2026

- Engineered a ROS 2 autonomous navigation pipeline integrating LiDAR-based SLAM, AMCL localization, and Nav2 path planning for real-time mapping and navigation.
- Tuned SLAM, costmaps, and planner parameters in Gazebo/RViz to improve map consistency and navigation stability in cluttered environments.

Human-Robot Interaction System | ROS 2, ESP32, micro-ROS, Python | [GitHub](#)

2025

- Built a real-time human-robot interaction pipeline connecting ESP32 hardware to ROS 2 through micro-ROS for low-latency gesture-driven control.
- Improved reliability by debugging serial communication, timing bottlenecks, and ROS 2 node interactions under embedded hardware constraints.

Soil Grain Detection & Mapping System | YOLOv8, OpenCV, Raspberry Pi, Folium | [GitHub](#)

2025

- Developed a YOLOv8-based soil grain classification pipeline using a custom dataset, achieving 92% detection accuracy across multiple soil categories.
- Implemented real-time inference in Python/OpenCV with edge deployment considerations for Raspberry Pi and GPS-based mapping outputs.

PID Line-Following Robot | Arduino, C, PID Control, CAD | [GitHub](#)

2024

- Designed and built a high-speed line-following robot using a C-based PID controller for smoother and more accurate tracking than basic threshold control.
- Integrated sensor array hardware, motor control logic, and chassis design to improve response, stability, and corner handling.

TeleOperated Mobile Manipulation Platform | ESP32, 4-DOF Kinematics, Bluetooth

2023

- Developed a teleoperated mobile robot with a 4-DOF robotic arm using forward/inverse kinematics and Bluetooth-based real-time control.

EXPERIENCE & LEADERSHIP

Project Committee Member — Embedded Systems Club

Aug 2024 – Sep 2025

New Horizon College of Engineering

Bengaluru, Karnataka

- Organized workshops on embedded systems for student communities and supported hands-on learning around microcontrollers and embedded development.
- Collaborated on club projects involving sensor integration, circuit design, and embedded programming, strengthening cross-functional teamwork and prototyping skills.

CERTIFICATIONS

- NVIDIA Deep Learning Institute — AI on Jetson Nano
- AWS — Introduction to Robotics
- IBM — Artificial Intelligence Fundamentals
- Universal Robots — Industrial Robot e-Learning